

GENERAL SPECIFICATION G7.4

Fire Protection, Detection and Alarm System

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G7.4 FIRE PROTECTION, DETECTION AND ALARM SYSTEM

G7.4.1 General

- (a) The Contractor shall be responsible for the detailed design of the fire protection, and alarm system for all site areas as indicated on the Drawings and as specified.
- (b) All firefighting /protection and alarm system shall be of a type approved by the Singapore Authorities and listed with either FM or UL agencies.
- (c) The Contractor shall submit to the S.O. for approval, details of the proposed systems.

G7.4.2 Reference Standards

- (a) The above works shall comply with the relevant Reference Standards listed below:

Standard	Subject
Fire Code 2013	Code of Practice for Fire Precautions in Buildings
SS CP 10	Code of Practice for installation and servicing of electrical fire alarm systems
SS 575	Code of Practice for fire hydrant systems, rising mains and hose reels
SS 578	Code of Practice for use and maintenance of Portable fire extinguishers
SS EN 3-7	Specification for Portable Fire Extinguishers
SS CP5	Code of Practice for Electrical Installation
BS EN671-1	Hose Reels for fixed installations
BS 6207	Specification for mineral-insulated copper sheathed cables with copper conductors
BS 5499	Fire Safety Signs. Notices and graphic symbol
BS 5306	Fire Extinguishing installations and equipment on premises - Hydrant Systems, Hose Reels and Fire Extinguishers

- (b) Should the requirements of the Drawings or Specification conflict with the codes, regulations or standards of the relevant authorities, the most stringent of the requirements shall govern the Works.

G7.4.3 Approval of Drawings and Details

- (a) In addition to the requirement set out elsewhere in the Contract, the Contractor shall submit the following:
 - (i) Circuit diagrams of main fire and sub-fire alarm panel and their components;

- (ii) Drawings showing the location of firefighting equipment, water distribution mains, hose reels, extinguishers, fire alarm detectors and manual call points, breakglass and strobe lights, and indicating equipment, supports, and wiring routes;
 - (iii) General arrangement drawings for control and alarm panels and fire protection equipment;
 - (iv) Circuit diagram of battery charger and its component;
 - (v) Layout and construction of main fire and sub-fire alarm panels including the battery compartment;
 - (vi) Technical specifications, Manufacturer's catalogues, data sheets and samples (if required) of all equipment employed in the installation;
 - (vii) Certificates for tests carried out at Site. Other information of the system as specified in this Specifications or required by the S.O.; and
 - (viii) All necessary calculation to determine the system pressure and head for achieving the required performance and testing.
- (b) Work shall only commence upon approval of the Contractor's proposal by the S.O.. The approval shall in no way relieve the Contractor of his responsibility should the plant or any of its parts proved inadequate in any aspect.

G7.4.4 As Built Drawings

- (a) In addition to the requirements set out elsewhere in the Contract, on completion of the installation, the Contractor shall provide the "As Built" drawings of the whole fire alarm protection installation and documents incorporating but not limited to the following information:
- (i) Positions and boundaries of various alarm zones groups;
 - (ii) Locations of every manual call point, alarm bell, fire and sub-fire board, battery and charger assembly;
 - (iii) Schematic circuit diagram and single line diagram of the overall system wiring;
 - (iv) Legend showing clearly all the symbols and abbreviations used;
 - (v) Circuit diagram of the main fire and sub-fire alarm panel and its components;
 - (vi) Circuit diagram of the battery charger and its components;
 - (vii) Diagram showing the external and internal layout of the main fire;
 - (viii) Alarm panel, sub-fire alarm panel and repeater panel;
 - (ix) Routes of all hose reel risers and pipes;
 - (x) Locations of every hose reel and portable fire extinguisher; and
 - (xi) Hydrant system, including location and supply network.

G7.4.5 Fire Alarm Systems

- (a) The fire alarm system shall be a closed circuit continuously-monitored installation, comprising essentially of a main fire alarm panel, sub alarm panels, triggering devices and alarm bells, fed from a continuously charged DC power supply source.

(b) Fire Alarm

- (i) The operation of one or more trigger devices shall result in an immediate fire alarm given by:
 - An audible alarm from all the alarm bell(s);
 - A visible indication for each zone in which a trigger device operates; and
 - A visual alarm from the strobe light attached and situated at each respective building.
- (ii) Where a number of alarm signals are received in quick succession, or simultaneously, the equipment shall be designed so that the audible alarm continues or resounds as each visible indication commences.
- (iii) The alarm state shall be maintained by the main fire alarm panel without a continued signal from the trigger device so that even destruction of a connection after the trigger device operates shall not discontinue the sounding of the alarm.
- (iv) The audible and visual alarms shall continue until silenced by the manual operation of a silencing switch; they shall not be automatically silenced. The operation of a silencing switch while an alarm condition exists shall not cancel the visual indication of the fire alarm nor shall the operation of silencing switch prevent the proper receipt of alarms from any zone(s) not yet providing an alarm signal. In the event of operation of a trigger device in any zone in which an alarm has not been received and silenced, further fire alarms as specified above shall be given.
- (v) There shall be only one switch for the silencing of the audible and visual fire alarm. In addition, the operation of the audible and visual fire alarm resounding system shall not require the activation of any other manual switches and devices.
- (vi) The restoration of the silencing switch to its normal position whilst an alarm condition exists shall transfer the alarm to the fire alarm bells.

(c) Fault Warning

- (i) The fire alarm system shall be capable of detecting fault automatically whenever it develops in any of the circuits and a fault warning shall be given by both an audible warning from a buzzer and a visible indication in the main fire alarm panel. This buzzer in the main fire alarm panel shall be distinctive and of a different character from fire alarm bells.
- (ii) The buzzer warning shall be cancelled by a single silencing switch solely designed for this purpose and unless the fault is rectified, the visual indication shall persist. The removal of fault shall automatically reset the audible fault warning circuit. The occurrence of another further fault while the buzzer is silenced shall cause the buzzer to sound again. Similarly, the restoration of the silencing switch to its normal position whilst a fault exists shall transfer the fault back to the audible fault warning of the system.
- (iii) Any fault warning shall not prevent a fire alarm being given at the same time. At any time, an alarm of fire shall take priority over any other indication such as fault warning.

- (iv) A fault warning as required shall be given in the event of any of the following situations:
- Failure or disconnection of the normal power supply;
 - Failure such as low voltage or disconnection of the standby battery;
 - Failure or disconnection of the battery charger;
 - Triggering circuit faults such as open circuit, short circuit, earth fault of the +ve or -ve line in the wiring between trigger devices and the main fire alarm panel including the removal of any trigger devices;
 - Bell circuit faults such as open circuit, short circuit, earth fault of the +ve or -ve line in the wiring between alarm bells and the main fire alarm panel including the removal of any bells; and
 - Rupture or disconnection of any fuse or the operation of any protective devices that would prevent a fire alarm being given.
- (d) Main Fire Alarm Panel
- (i) The main fire alarm panel shall be a solid state electronic type and shall be located as shown in the drawings. It shall contain the following equipment:
- 1) Switches and indicating facilities for zones;
 - 2) Individual reset switches per indicator;
 - 3) "Mains On", "Charger On" and "System On" indicator;
 - 4) "Mains Fault", "Charger Fault" and "System Fault" indicators;
 - 5) Battery test facilities with voltmeter and centre zero ammeter (to indicate charging and discharging current of the battery);
 - 6) Ammeter to indicate charger output current;
 - 7) Battery test switch;
 - 8) Mains "On" indicator;
 - 9) Silencing and isolating switch for alarm bell/strobe light;
 - 10) Indicator for alarm bell isolation;
 - 11) Audible fault warning buzzer;
 - 12) Silencing switch for fault warning buzzer;
 - 13) Indicator for buzzer isolation;
 - 14) Indicator for bell circuit fault;
 - 15) Battery low voltage and disconnected indicator;
 - 16) Alarm bell/strobe light test push button;
 - 17) System reset switch; and
 - 18) Auxiliary contacts.

- (ii) The fire alarm board shall be accommodated in the well-finished self-sustained sheet cabinet constructed from at least 1.5 mm thick hot dipped galvanised sheet steel. The cabinet housing shall be bright red in colour and shall be fitted with a hinged, glazed door which shall be locked. All alarm zone indicators and associated switches shall be visible behind the locked door. The manufacturer's name shall be clearly and permanently marked on the front of the cabinet.
- (iii) The compartment housing the batteries and the charger shall be totally enclosed and fitted with vermin proof louvres at the front plate. The internal shall be fully lined with Formica or other approved materials against corrosive elements from the batteries.
- (iv) Spare zones shall be provided for the main fire alarm panel, 5 for those with 10 or more zones and 2 for those with less than 10 zones. Facility shall be provided in the main fire alarm panel so that easy connection to extension unit can be made when future expansion of the main fire alarm panel becomes necessary.
- (v) A manual call point and an alarm bell/strobe light shall be provided adjacent to the main fire alarm panel.
- (vi) All main fire alarm panels shall incorporate the automatic facilities to receive, interpret and operate the signals from their firefighting systems installed in the buildings.
- (vii) Black Perspex labels with white engraved lettering designating the alarm zone, the functions of the various indicators, apparatus and switches shall be provided for every fire indicator board. All manual controls shall be clearly labelled to indicate their functions. An information chart listing the circuit schedule shall be fixed inside each board. Other type of labelling may also be used subject to the approval of the S.O..
- (viii) Approved silk screen white Perspex sheet with reduced scale coloured drawings of approved sizes showing the boundaries and locations of the alarm zone shall be provided by the Contractor and located next to each fire indicator board, sub-fire indicator board and repeater board.
- (ix) Locations of fire protection equipment such as fire indicator board, portable fire extinguishers, fire hose reels, dry riser, etc. shall also be indicated on the white Perspex coloured drawing in the form of clear and distinctive symbols.
- (x) Escape routes including corridor, staircase, etc. shall be clearly indicated in the above zoning diagram.
- (xi) An auxiliary contact shall be connected to the hardwired Alarm System in the control room.
- (e) Alarm Zone - Zone Switches and Indicating Facilities
 - (i) The following switches and indicating facilities shall be provided for each alarm zone or group:
 - 1) Fire alarm indicator;
 - 2) Fault indicator;
 - 3) Isolation indicator;
 - 4) Alarm test switch;

- 5) Fault test switch;
 - 6) Normal position switch; and
 - 7) Isolation position switch.
- (f) Fire, Fault and Isolation Indicators
- (i) Each indicator shall be a twin light emitting diodes (LED) and shall light up upon actuation and remaining lit until manually reset.
 - (ii) The fire alarm indicator shall be coloured red and clearly labelled with the word "FIRE"; fault warning indicator shall be coloured amber and clearly labelled with the word "FAULT" and zone isolation indicator shall be coloured green and clearly labelled with the word "ISOLATED".
- (g) Alarm Test Switch
- (i) Test facilities to simulate operation of a trigger device without transmitting of a fire alarm signal shall be provided for each alarm zone. Provision shall be made so that the operation of the remainder of the system is not impaired while testing of a particular circuit is carried out.
- (h) Fault Test Switch
- (i) Test facilities to produce a circuit fault condition in the supervisory circuit of the wiring to the remote actuating devices shall be provided in the main fire alarm panel for each alarm zone or group.
- (i) Isolation Switch
- (i) A switch shall be provided for the deliberate isolation of each individual alarm group from the main fire alarm panel common circuitry. Operation of the isolation switch shall prevent the alarm group from initiating either alarm or fault signals and shall not impair the normal functioning of any other alarm groups.
- (j) Indicators
- (i) Lamp indicators shall be the light emitting diode (LED) type.
 - (ii) The arrangement of indicator circuits and the design of the apparatus shall be such that the operation of one indicator does not prevent the proper and separate operation of other indicators.
 - (iii) The operation of any alarm bell or the transmission of fire alarm signal to Monitoring Station shall not be prevented by any faulty indicator and shall not depend on the operation of any indicator.
- (k) Battery Test Facilities
- (i) A non-locking self-resetting battery test switch shall be provided to apply a 300% load condition by the use of resistor bank to the battery with the battery charger disconnected or switched off.
 - (ii) An appropriate rated direct current ammeter shall be fitted to the main fire alarm panel to indicate the loading current during the above battery test.
 - (iii) A direct current voltmeter, scaled 0V to 48V, shall be provided at the fire alarm panel to monitor the terminal voltage of the battery.

- (l) Connection of Ancillary Services
 - (i) Ancillary services shall not affect the functioning of the equipment.
 - 1) A 24V DC supply may be provided from the fire alarm control panel to operate a relay or similar operating device controlling the ancillary equipment. The mechanism shall be such that it does not impose a continuous load on the battery under non-alarm conditions. Provision for the interface between the main fire alarm panel and auxiliary plant, control equipment or external devices which require to be operated or disconnected in the event of an alarm shall also be made by the use of at least two separate and uncommitted relays. Each relay shall provide 2 sets of voltage free changeover contacts having minimum rating of 5A 30V d.c.
 - 2) Outputs to any ancillary services shall be fused or similarly protected to ensure the safety of the control equipment.
- (m) Sub-Fire Alarm Panel
 - (i) The construction and other requirements of sub-fire alarm panels shall be similar to those for the main fire alarm panel. Sub-fire alarm panels shall contain indicators together with associated switches and indicating facilities for zones. The sub-fire alarm panel shall have additional facilities for transmitting a sector signal to the main fire alarm panel.
 - (ii) Where any sub-fire alarm panel is provided in a sector of a premises, the sector in which an alarm originates shall be shown on the main fire alarm panel, and the zone of origin of the alarm shall be shown on the sub-fire alarm panel. Where both sector and zone indication are provided on one main fire alarm panel, clear differentiation shall be made between the two functions.
- (n) Power Supply
 - (i) The fire alarm system shall be operated on a DC power supply of 24 V provided by sealed type maintenance-free lead-acid batteries or nickel cadmium which shall be designed to be maintained in an adequately charged condition by a 230 V AC/24 V DC rectifier/charger.
 - (ii) A double-pole 230 V rated switch with neon indicator shall be provided for the termination of the incoming mains supply. This supply shall be exclusive to the fire alarm system and connected to a separate circuit from the main switchboard and the controlling circuit breaker shall be clearly labelled "FIRE ALARM: DO NOT SWITCH OFF".
 - (iii) The batteries shall be suitable for standing on open circuit or floating across a constant voltage supply and discharging at infrequent intervals.
 - (iv) The capacity of the battery used to energise the alarm system shall be such that in the event of mains failure, the battery shall be capable of maintaining the system in normal working condition for at least 72 hours. Thereafter, it shall be capable of supplying an additional load resulting from an alarm originating in two separate alarm zones or groups for a period of half an hour and, if utilised to supply emergency evacuation alarms, it shall in addition be capable of supplying the full emergency evacuation alarm load for a period of at least 10 minutes.

- (v) Calculations for the sizing of the batteries and the design of the battery charging equipment to meet the above requirements shall be submitted to the S.O. before the Contractor provides the equipment.
- (vi) The battery terminal pillars and inter-cell connectors, bolts, nuts and washers used for fastenings of inter-cell and terminal connectors shall be made of highly conductive and corrosion-free material.
- (vii) The cell container of the battery shall be enclosed and constructed of rugged and self-extinguishing type plastic.
- (viii) The battery shall be arranged in the compartment in such a way as to allow easy maintenance so that the inspection of batteries shall not require the removal of cells and connections from the battery assembly.
- (ix) A double-pole isolation switch shall be provided so that the battery can be isolated from the charger. The fault buzzer and an indicating light on the front panel of the main fire alarm panel shall be activated to announce the isolation.
- (x) A rating plate containing the following information shall be securely fixed within the battery enclosure:
 - 1) Type of cell;
 - 2) Numbers of cells;
 - 3) Ampere hour capacity;
 - 4) Nominal voltage;
 - 5) Final voltage;
 - 6) Float voltage of battery charger;
 - 7) Output current and limit of battery charger; and
 - 8) Date supplied and recommended date of replacement.
- (xi) Battery Charger
 - 1) The battery charger shall be of the constant voltage current limiting type with automatic boost charge by voltage level sensing and capable to function under the following environmental conditions:
 - Temperature: 10°C - 40°C;
 - Relative humidity: 20% - 95%; and
 - Abnormal foreign potential: Power line surge.
 - 2) Temperature compensated circuit shall be employed to uphold any changes in the charger under the specified operating temperature range.
 - 3) The charger shall be capable of supplying any possible load of the system while maintaining the battery in fully charged condition and recharging the battery as prescribed below. The charging system shall be current limiting constant voltage type.

- 4) The rating of the battery charger shall be such that:
 - After the battery has been discharged from the fully charged state by supplying the load as specified in the previous Clause; and
 - After recharging for a period of not more than 24 hours, the battery shall have recovered to the extent that it shall be capable of sustaining a further discharge of supplying the same load as specified in the previous Clause.
- 5) The output voltage provided by the battery charger shall be maintained within 0.1% of the pre-set value whilst operating within any combination of the conditions described below and whilst delivering a current within the range of values required for float charging:
 - Voltage: 230V r.m.s. +10%; and
 - Frequency: 50 Hz + 5% single phase
- 6) The battery charger shall keep the battery under constant voltage charge. It shall incorporate automatic control features with output designed to charge and maintain the cells of the battery within the limits specified by the battery manufacturer, taking into account any quiescent load imposed by the associated system.
- 7) The voltage output of the charger shall be adjustable to +20% of the nominal voltage. The current limit shall be adjustable to +20% of the maximum allowable charging current of the battery. The trimmers shall be arranged so that adjustment can be done without removing any circuit boards or obstacles.
- 8) All electronic parts including rectifiers, power transistors and resistor shall be designed to function at the case temperature below 75°C unless otherwise approved by the S.O..
- 9) Printed circuit boards, fuses, relays and other electrical parts employed in the battery charger shall be mounted and connected in such a way that inspection of them does not require them to be dismantled from their operating positions.
- 10) An approved self-resetting low voltage cut-off device with adjustable low voltage setting shall be installed in the circuit between the battery and the system.
- 11) This under-voltage protection device shall disconnect the battery from supplying the system if the pre-set low voltage limit as recommended by the battery manufacturer is reached.

G7.4.6 Manual Call Points

- (a) Manual call points shall be of the "break-glass" type complying with the requirements of EN 54: Part 11, and notwithstanding this, shall be of a type approved by the local governing authority.
- (b) The casing of the call point shall be of shatterproof corrosion resistant plastic, finished red in colour. The front glass fitted shall be replaceable and shall be such that the possibility of accidental breaking is reduced to a minimum.

- (c) Manual call points installed outdoors shall be of approved weatherproof and watertight steel box construction complete with screwed conduit entry hole.
- (d) The letter "FIRE" shall be embossed on the front surface.
- (e) Manual call points shall be mounted at a height of 1.4m above finished floor level.

G7.4.7 Automatic Fire Detectors

- (a) Automatic fire detectors shall be of the types approved by the statutory authorities. The detectors shall preferably be flush mounted in offices but surface mounted in plant rooms and shall be installed in such a manner that each can be readily removed for servicing without affecting the effectiveness of the remaining detectors and the system operation.
- (b) Smoke detectors shall be of the ionization or photoelectric type as specified.
- (c) Heat detectors shall be of combined rate-of-rise / fixed temperature heat detector type as specified.
- (d) Detectors shall have light emitting type operation indication and shall be suitable for fixing direct to circular conduit boxes.

G7.4.8 Alarm Bells and Strobe Light

- (a) Alarm bells shall be of the cast or pressed steel under-dome type, finished red in colour. They shall be complete with a strobe light on top of the bell cover. The bell shall be the trembling type with a gong diameter of 150mm, and shall be capable of giving a minimum sound pressure level of 90 dB(A) within a range of 1m continuously.
- (b) The strobe light shall have a high intensity flashing strobe light with flash rate of 50 to 80 flashes per minute. The lens cover of the light shall be red in colour. Alarms shall be installed above near entrance/exit doorways near to the audible alarm.
- (c) The bells and light shall be suitable for operation from a 24 V DC power supply.
- (d) Alarm bells shall be mounted on conduit boxes at ceiling height.
- (e) Alarm bells installed outdoors shall be of approved weatherproof and watertight construction. Approved special sealing gaskets or gasket-sealed boxes shall be used.

G7.4.9 Cabling Installation

- (a) Openings for cables to pass through structural fire barriers shall be sealed with an equal degree of fire resistance as the structural member. Channels and ducts carrying cables shall be sealed internally under these circumstances.
- (b) Cabling for the fire alarm system shall be installed in accordance with the appropriate provisions of BS and local authorities' standard.
- (c) The cabling for the fire alarm system and its associated extra-low voltage circuits shall be exclusive to the fire alarm installation and shall be segregated from the cabling of any other circuit. Fire alarm cabling shall be spaced at least 50mm away from any other service. The mains feed to the power supply equipment of the fire alarm system shall be segregated from the cabling of all other services and also from all other circuits of the fire alarm system.

- (d) Each alarm zone or group shall be separately wired and indicated on main fire alarm panel.
- (e) The conductor of the cable shall have suitable cross-sectional area of not less than 1.5 mm² and shall be multi-stranded.
- (f) All cabling except the low voltage mains feed to the power supply equipment of the fire alarm system shall be fire resistant cable. Fire resistant cable shall be a single or multi-core unarmoured cables rated at 300/500V, comprising plain copper stranded circular conductors, fire barrier tape and insulated with extruded polythene (PE), or ethylene propylene rubber (EPR), or other approved fire resistant compound, and an extruded over sheath of PVC. Fire resistant cables shall pass the type testing conducted by local approved testing agency. All cabling shall be run in steel metal conduits or metal trunking exclusive to the fire alarm system. The metal conduits shall be painted red and shall be suitably labelled.
- (g) Manual call points shall be connected in an "open" circuit and an "end-offline" resistor shall be provided to enable line monitoring.
- (h) No joints between terminal points shall be permitted except in long runs, in which case the joints shall only be made in a suitable enclosed terminal box employing screwed terminations. The box shall be labelled "Fire Alarm Link Box".
- (i) The cabling diagram shall be submitted to the S.O. for approval before installation commences.
- (j) The alarm bell circuits shall be so arranged that a short circuit or disconnection of a single point in the cabling does not result in the silencing of all alarm bells in the same circuit. An appropriate end of line resistor shall be fitted to the terminals of the last alarm bell for line monitoring. All alarm bells shall be polarized and suppressed.

G7.4.10 Hose Reels

- (a) Hose reels shall be either fire hose reels or wash-down hose reels as shown on the Drawings. Fire hose reels shall be clearly marked as such and shall be painted red. Wash-down hose reels shall be painted the same colour as the potable water pipework and shall be clearly marked.
- (b) Fire hose reels shall be of the recess double swivel swing type and comply with all regulations and requirements of the Local Authorities.
- (c) Where a hose reel is not recessed, it shall be housed in a presentable metal cabinet to the S.O.'s approval.
- (d) The hose reel shall extend not more than 350mm from the mounting surface and shall have an overall diameter not greater than 760 mm. The side plates of the hose reel shall have a continuous outer rim and shall be rigid in construction. The side of the reel shall be painted in red.
- (e) The diameter of the inner core on which the hose is wound shall be not less than 200 mm. When the full length of the hose is coiled on the hose reel without the use of abnormal tension, the rim of the side plates of the hose reel shall project at least 10 mm beyond the external diameter of the coiled hose of any point.

- (f) The fittings to which the hose are attached on the hose reel shall be arranged in such a manner that the hose is not restricted or flattened by the application of the additional layers.

(i) Markings

- 1) Every hose reel shall be marked with the following information:
 - The Manufacturer's name, trade name or mark.
 - Instructions for operation and use which shall include the following:
 - Turn on stop valves or release nozzle;
 - Run out hose;
 - Turn on water at nozzle;
 - The hose shall be fully discharged before winding it onto the reel;
 - The year of manufacture; and
 - The test pressure in bar.
- 2) The Contractor shall fix the hose reel so that the centre is between 0.75 m and 1.4 m above floor level and shall mount the hose reel on a swivel bracket suitable for mounting in a 1.6mm thick sheet steel hose reel cabinet or recesses clearly marked with "FIRE HOSEREEL" in white letters of at least 50 mm high. Hose reel cabinet and recess shall be painted red.
- 3) Any doors provided for hose reel cabinet or recesses shall be hinged so that they open approximately 180 degrees and they shall not obstruct the running out of hose in either direction. The door shall be locked and the master key shall be placed inside a recess provided with front break glass on each door.
- 4) The Contractor shall provide five (5) sets of master keys to the Board.
- 5) The Contractor shall arrange with approved testing local authority/agent for a test of the hose reels and shall submit test reports from the local authority/agent to the S.O. before delivery of the hose reel to the Site. All costs required for such test carried out by local authority / agent shall be borne by the Contractor.
- 6) Notwithstanding the above requirements, hose reels shall be factory tested against water leakage after assembly at a test pressure of 13.8 bars (200psi) for a duration of 30 seconds for each unit and the Contractor shall furnish documentary evidence of such factory test to the SO before delivery of hose reels to Site.

(g) Rubber Hose

- (i) The rubber hose shall be of 25 mm internal diameter and shall have a minimum working pressure of 7 bars and shall be of the non-kinking type.
- (ii) The rubber hose shall be manufactured with inner tube or lining, a reinforcement braided with textile material and an abrasion resistant rubber cover. The reinforcement shall consist of a single rayon braid or double braid of cotton.

- (iii) The length of the rubber hose on the reel shall not be less than 30 meters.
- (iv) There shall be no joints in the length of hose.

(h) Shut-off Nozzle

The Contractor shall provide a shut-off nozzle assembly in accordance with the following requirements:

- (i) The assembly shall be constructed of a suitable corrosion-resistant metallic material.
- (ii) Means shall be provided for the attachment of the assembly to the top valve in such a manner that it cannot be disengaged until the stop valve has been opened.
- (iii) The assembly shall be permanently marked to indicate the open and shut position of the valve.
- (iv) The internal surface of the nozzle shall be designed to provide a smooth surface.
- (v) The internal diameter of the outlet shall be 6.35mm.

(i) Pipes and Fittings

- (i) The Contractor shall provide all pipework required for correct operation of the system and for compliance with the Code of practice and standards.
- (ii) Fire hose reel pipework and risers shall be of 50 mm nominal diameter throughout except the individual feed pipe to each hose reel which shall be of 25 mm nominal diameter.
- (iii) Where necessary, the Contractor shall lay pipes outside buildings underground. The depth of cover measured from the top of capillary fittings and compression fittings shall be in accordance with BS EN 1254-2. Fitting of 15 mm and longer shall be legibly marked with maker's name or trade mark. Heavy galvanised steel pipe shall comply with BS EN 10255 or equivalent.
- (iv) Joints and connections to galvanized steel pipework of up to and including 65mm diameter shall be by means of screwed connections. Pipes of 80mm and above shall be joined by means of flanged connections. Unless otherwise specified, galvanized steel bolts, nuts and washers shall be provided for all supports and flanged connection.
- (v) The minimum nominal pipe diameter to be used for all drain and air vent connections shall be 20 mm. All drain pipes shall be terminated visibly and at suitable location for drainage purposes.
- (vi) All conduits, hangers, supports, equipment, trunking, pipes and other work shall be cleaned and given one primary coat and two finishing coats of approved red colour enamel paint unless otherwise directed by the S.O..

- (j) Valves
 - (i) In addition to the specification set elsewhere, the Contractor shall provide 25 mm diameter stop valve for the connection of the hose reel to water supply. An indication of the open and shut position shall be fixed or permanently marked on the wheel of the valve. The body of the valve except for gate valves shall be marked with a directional arrow indicating the direction of flow through the valve. The water connection through the rubber hose shall permit full flow of water to the hose without external leakage during rotation of the hose reel. The hose reel waterways connecting the water supply to the hose, excluding the stop valve and that part between the stop valve and the hose reel shall be constructed of non-ferrous material and / or stainless steel.
 - (ii) All stop valves shall be 'right-handed' i.e. they shall be so constructed in order to shut the valve, the spindle shall turn clockwise. The hand wheels shall be clearly marked showing the direction the wheel shall be turned to close the valve.
- (k) Vents
 - (i) The Contractor shall provide automatic air vent of approved manufacture at the top point of each pipe loop of a piping system and at the top point of each pressurised pipe stack.
 - (ii) Where a wet rising main is above the water entry into it, it shall also be equipped with an automatic air release valve at the top.
- (l) Pipe Supports and Fittings.

Refer to section G4 for General Specification.

G7.4.11 Fire Sprinklers

- (a) Refer to General Specification G7.15 – Sprinkler System.

G7.4.12 Dry Rising Main

- (a) The Contractor shall supply, deliver and install wet and dry rising mains as shown in the drawings.
- (b) The rising main installed shall comply with all regulations and requirements of the Singapore Standard SS 575: Code of Practice for Fire Hydrant, Rising Main and Hose Reel Systems
- (c) The materials used for rising mains/down comers shall be of hot dipped galvanised steel pipe in accordance with BS EN 10255 and heavy quality; fittings shall be of wrought iron steel.
- (d) Air relief valves shall be provided at the highest point of each dry rising mains to permit air to be discharged to atmosphere when water is pumped in at ground level.
- (e) The down comers shall have pressure relief valves to limit the pressure and also shall arrangements made in accordance with BS 5041: Part 1 so that when the water is shut off at the nozzle, the static pressure in any line of hose connected to a landing valve does not exceed 7 bar.

- (f) Each rising main has to be connected to a 2 way or 4 way breeching inlet with instantaneous male couplings for connecting to the Fire Authority's 63.5mm diameter standard hose. They are to conform to BS 5041: Part 3. At each base of the rising main 25mm diameter drain valve is to be provided.
- (g) Each rising mains shall be bonded to earth and mechanical groove end pipe couplings are used, ensure continuity conductor is installed across each joint.
- (h) Standby hose assemblies house in metal casing enclosure shall be provided for each landing valve location.

G7.4.13 Fire Extinguishers

- (a) The Contractor shall supply, deliver to site and install fire extinguishers in compliance with Fire Codes and Regulations.
- (b) The use of halon gas in fire protection system will not be permitted. Suitable alternatives shall be used in compliance with SS 578.
- (c) Each fire extinguisher shall have a tag or label securely attached that indicates the date of inspection and maintenance was performed and shall identify the person and organisation performing the service. The same record tag or label shall indicate if recharging was also performed and when the next pressure test is to be conducted.
- (d) Fire extinguishers shall be installed at a maximum height of 1 m above the finished floor level on the hangers / brackets and shall comply with SS EN3-7. They shall be placed in a manner such that their operating instructions face outward.
- (e) The operating instructions located in front of the fire extinguisher shall be in accordance with SS 578 and SS EN3-7.
- (f) The fire extinguishers shall not suffer from corrosion at the interface between the cylinder body and nozzle assembly due to galvanic action between dissimilar metals.
- (g) Each Extinguisher (except those in service rooms, substation) shall be housed in a metal box complete with a front-hinged door, 2mm thick glass panel, handle and master lock. The case shall be coated with 2 coats of red paint. Lettering such as "FIRE" or "FIRE EXTINGUISHERS", shall be eminently and permanently marked on the case and shall be of approved size and colour. The contractor shall supply 5 sets of Master Keys to the S.O..
- (h) Extinguishers inside service rooms electrical substation do not require metal cases. Their holders / brackets shall be gun tooled firmly onto the walls/supporting surfaces.

G7.4.14 Fire Hydrant

- (a) The Contractor shall supply, deliver and install fire hydrants as shown in the drawings.
- (b) The fire hydrant installed shall comply with all regulations and requirements of the SCDF.

- (c) Hydrant supply pipe laid underground shall comply with the following:
 - (i) Steel spigot and socket pipe to SS 142; and
 - (ii) Grey iron pipe with factory applied sulphate resisting cement lining. Pipes to be coated externally with metallic zinc followed by bituminous finish coat. Bolts, nuts and studs to be hot dip galvanised and nylon coated.
- (d) 150mm (NB) diameter main pipe shall be installed for the connection to the Board's main supplying water to the fire hydrant.
- (e) Double pillar provided shall c/w 2 outlets each 63.5mm suitable for connection to the local fire brigade equipment.
- (f) Hydrant pipe buried shall be 1m below ground wrap round with 100mm of sand bed, anchor blocks shall be provided at the pipe joints.
- (g) Valve pit shall be constructed at the hydrant valve location either in brick or pre-cast concrete as per details in SS CP 29.
- (h) Private fire hydrant shall have 100mm wide yellow band painted on stem with lower edge 300mm from finished ground level. Hydrant marking plate shall be in accordance to BS 3251.
- (i) Water test shall be conducted at 13.8 bar (200 psi) pressure for 2 hours or 3.5 bar (50 psi) in 2 hours in excess of the maximum pressure where the maximum pressure exceeds 10.3 bar (150 psi)

G7.4.15 Inspection, Testing and Commissioning

- (a) In addition to the specification set elsewhere, before the building handover date to the Board all defects are to be properly rectified.
- (b) Provide a complete set of testing equipment. The minimum requirements of which shall be set out below:
 - (i) Meter-recorder for measuring flow rate; and
 - (ii) All necessary testing equipment such as insulation resistance tester.
- (c) Fire Alarm System Site Testing
 - (i) The entire system shall be tested to ensure that it operates satisfactorily and that:
 - 1) All trigger devices and alarm points function correctly;
 - 2) All fault monitoring circuits function correctly;
 - 3) The battery and battery charger comply with the requirements of this Section;
 - 4) All ancillary equipment functions correctly; and
 - 5) All control and indicating features provided in the main fire alarm panel and sub-fire alarm panel function correctly.

- (d) Pipework, Testing and Cleaning
 - (i) In addition to the requirement set elsewhere in the specification. All pipework and other fittings shall be hydrostatically tested to 10 bars and two times the working pressure whichever is the greater before lagging or putting into service. The test pressure shall be maintained for a minimum duration of 24 hours. Allowable pressure drops during 24 hours testing shall be decided by the S.O. but generally shall not exceed 3% of test pressure. During tests, all welded joints shall be hammered.
- (e) Hose Reel Tests
 - (i) Before any water is introduced into the hose reel, it shall be thoroughly flushed out to ensure that no harmful matter is passed into the reel. After flushing out, the tubing shall be run out and the water turned on to establish that the assembly is fully operational.
 - (ii) The two topmost and hydraulically remote hose reels shall be discharged simultaneously, one being led to waste and the other directed into a suitable receptacle to ascertain that the flow rate of at least 0.5 litres per second and 6 metres in horizontal throw of the jet from each reel have been achieved.
 - (iii) When the tests have been satisfactorily completed, restore the system to its normal operating state. Reels shall never be left under pressure and, where practicable, drain the hose prior to returning it to the drum.
 - (iv) Restore the hose on the drum, pay particular attention to automatic hose reels to ensure that the automatic valve is fully closed when the hose has been restored. Where an isolation valve has been incorporated, close the isolation valve on completion of storage and complete the nozzle interlock, where fitted.

G7.4.16 Service and Maintenance

- (a) In addition to the Specification set elsewhere, at each such monthly inspection and service of the complete plant and ancillary equipment, the minimum items of work detailed below shall be performed by the Contractor. Any special maintenance requirements on equipment not covered by the following list but recommended by the Manufacturers shall be included:
 - (i) Check all bolts and nuts for tightness, and tighten as necessary;
 - (ii) Inspect all hose reel;
 - (iii) Ensure that the inlet valve, the automatic on/off valve (in any), glands, tubing, and shut-off nozzle are sound and free from leaks, and also to ensure that the outlet of the nozzle is not choked. Check hose reels and rubber hose for damage;
 - (iv) Some nozzles, in addition to giving a jet stream, are also capable of producing cone spray. In these cases, the correct functioning shall be checked;
 - (v) Inspect and check all riser pipes for water leakages and rectify as necessary;
 - (vi) Inspect and check all portable fire extinguishers that they are in their proper position and have not been discharged, or lost pressure (in the case of extinguisher fitted with a pressure indicator), or suffered obvious damage;

- (vii) Inspect, check and replenish as necessary all appropriate fire extinguishing agents, which are defective. All extinguisher nozzle shall be checked for blockage once in 6 months;
- (viii) Instruct the persons responsible for the operation of the plant and equipment on correct method of operating the plant and equipment, and on the maintenance points to be watched;
- (ix) Simulate fire conditions by operating test switches of all alarm groups and check to ensure all relevant indication/groups/alarm (audio and visual) are working. Resetting the installation to normal and confirmed tests have been completed;
- (x) Check proper operation of the changeover device and battery voltage by operating the battery test switch and record voltage under load conditions in the log book. Check also the charging current to verify that it is within the permissible limit. Check that the battery is operating within the normal float voltage limits marked on the scale of the DC voltmeter provided on the battery charger. Recording the reading in the log book. Check all connections for tightness and signs of corrosion;
- (xi) Check conditions of batteries including terminals, leakage and correct level of electrolyte in each cell to ensure that they are in good serviceable condition. Top up battery with ONLY the specific type of liquid as recommended by the manufacturer;
- (xii) Check condition of battery cabinet for corrosion and to ensure that batteries are stored in a secure condition. Remove any dust or dirt that may be accumulated within the cubicle;
- (xiii) Check to ensure that all indicating lights are operating correctly and replace if faulty;
- (xiv) Check operation of all alarm bells;
- (xv) Check power supply failure facility by disconnecting the battery and main supply;
- (xvi) Simulate fault condition on all alarm groups to ensure that the fire alarm panel is operating correctly including the fault warning buzzer;
- (xvii) Check the fire alarm panel is in a clean and operative condition and ensure that the enclosure is maintaining a satisfactory dust seal;
- (xviii) Visually inspect the condition of components, including resistors, capacitors and cable;
- (xix) Ensure that the faulty parts are replaced and the replacement has been carried out satisfactorily. Note in the log book any circuits that are subject to repair, alteration or extension; and
- (xx) Check all switches are returned to their normal operating condition.

G7.4.17 Annual Tests

- (a) In addition to the regular monthly inspection and service, the Contractor shall perform the annual servicing and maintenance as detailed below:
 - (i) All the monthly inspection and testing procedures specified above;
 - (ii) The hose shall be completely run out and subjected to operational water pressure to ensure that the hose is in good condition and that all couplings are water tight. A flow testing shall be carried out to ensure that a discharge of at least 0.5 l/s is achieved. If it is not possible to test every hose reel, at least the highest reel on each rising main shall be tested;
 - (iii) Replace any broken or faulty equipment as may be required;
 - (iv) Enter test results and details of any faulty components, repairs and replacements in the log book;
 - (v) Thoroughly inspect all extinguishers in accordance with the manufacturer's recommended procedure using the tools and equipment specified therein. Any extinguisher showing signs of corrosion or damage to the body shall be taken out of service and replaced. Faulty, damaged or corroded parts shall be replaced only by the correct component supplied by the extinguisher manufacturer;
 - (vi) The date of last inspection shall be indelibly recorded on a durable adhesive label firmly fixed to the extinguisher;
 - (vii) The annual maintenance schedule as specified in Clause 9 of BS 5306 Part 3 latest edition shall be carried out with respect to different types of extinguishers; and
 - (viii) The water in the whole hose reel pipework system shall be completely drained and filled with fresh water again.

G7.4.18 Log Book

- (a) Keep a log book at each separate building in which details of all alarms (genuine, practice test or false) faults, service tests and routine attention given shall be recorded. The exact location within each building where the log book is to be kept shall be subject to the approval of the S.O..

END OF SECTION